

在以往的学习中，我们接触过二元、三元等简单的线性方程组。

从许多实践或理论问题里导出的线性方程组常常含有相当多的未知量，并且未知量的个数与方程的个数也不一定相等。



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我们先讨论未知量的个数与方程的个数相等的特殊情形.

在讨论这一类线性方程组时, 我们引入行列式这个计算工具.



- 行列式是线性代数的一种工具!
- 学习行列式主要就是要能计算行列式的值.

内容提要

§ 1 二阶与三阶行列式

§ 2 全排列和对换

§ 3 n 阶行列式的定义

§ 4 行列式的性质

§ 5 行列式按行(列)展开

行列式的概念.

行列式的性质及计算.



§ 1 二阶与三阶行列式

我们从最简单的二元线性方程组出发，探求其求解公式，并设法化简此公式.

一、二阶行列式的引入

用消元法解二元线性方程组

$$\begin{cases} a_{11}x_1 + a_{12}x_2 = b_1, & (1) \\ a_{21}x_1 + a_{22}x_2 = b_2. & (2) \end{cases}$$

$$(1) \times a_{22} : \quad a_{11}a_{22}x_1 + a_{12}a_{22}x_2 = b_1a_{22},$$

$$(2) \times a_{12} : \quad a_{12}a_{21}x_1 + a_{12}a_{22}x_2 = b_2a_{12},$$

两式相减消去 x_2 , 得

请观察，此公式有何特点？

➤分母相同，由方程组的四个系数确定

➤分子、分母都是四个数分成两对相乘再相减

当 $a_{11}a_{22} - a_{12}a_{21} \neq 0$ 时，方程组的解为

$$x_1 = \frac{b_1 a_{22} - a_{12} b_2}{a_{11} a_{22} - a_{12} a_{21}}, \quad x_2 = \frac{a_{11} b_2 - b_1 a_{21}}{a_{11} a_{22} - a_{12} a_{21}}. \quad (3)$$

由方程组的四个系数确定。

$$\begin{cases} a_{11}x_1 + a_{12}x_2 = b_1, \\ a_{21}x_1 + a_{22}x_2 = b_2. \end{cases}$$

定义 由四个数排成二行二列（**横排称行、竖排称列**）的数表

$$\begin{array}{cc} a_{11} & a_{12} \\ a_{21} & a_{22} \end{array} \quad (4)$$

表达式 $a_{11}a_{22} - a_{12}a_{21}$ 称为数表 (4) 所确定的二阶

行列式，并记作 $\begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix}$ (5)
(**Determinant**)

即
$$D = \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} = a_{11}a_{22} - a_{12}a_{21}.$$

其中， a_{ij} ($i = 1, 2; j = 1, 2$) 称为**元素**。

i 为**行标**，表明元素位于第 i 行；

j 为**列标**，表明元素位于第 j 列。

二阶行列式的计算——对角线法则

主对角线

副对角线

$$\begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} = a_{11}a_{22} - a_{12}a_{21}.$$

对于二元线性方程组

$$\begin{cases} a_{11}x_1 + a_{12}x_2 = b_1, \\ a_{21}x_1 + a_{22}x_2 = b_2. \end{cases}$$

若记

系数行列式

$$D = \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix},$$

$$\begin{cases} a_{11}x_1 + a_{12}x_2 = b_1, \\ a_{21}x_1 + a_{22}x_2 = b_2, \end{cases}$$

$$D = \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix},$$

$$D = \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix},$$

$$\begin{cases} a_{11}x_1 + a_{12}x_2 = b_1, \\ a_{21}x_1 + a_{22}x_2 = b_2, \end{cases}$$

$$D = \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix},$$

$$D_1 = \begin{vmatrix} b_1 & a_{12} \\ b_2 & a_{22} \end{vmatrix},$$

$$\begin{cases} a_{11}x_1 + a_{12}x_2 = b_1, \\ a_{21}x_1 + a_{22}x_2 = b_2, \end{cases}$$



$$D = \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix},$$

$$\begin{cases} a_{11}x_1 + a_{12}x_2 = b_1, \\ a_{21}x_1 + a_{22}x_2 = b_2, \end{cases}$$

$$D = \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix},$$

$$D_1 = \begin{vmatrix} b_1 & a_{12} \\ b_2 & a_{22} \end{vmatrix},$$

$$\begin{cases} a_{11}x_1 + a_{12}x_2 = b_1, \\ a_{21}x_1 + a_{22}x_2 = b_2, \end{cases}$$

$$D_2 = \begin{vmatrix} a_{11} & b_1 \\ a_{21} & b_2 \end{vmatrix}.$$

则二元线性方程组的解为

$$x_1 = \frac{b_1 a_{22} - a_{12} b_2}{a_{11} a_{22} - a_{12} a_{21}} = \frac{D_1}{D}$$

$$x_2 = \frac{a_{11} b_2 - b_1 a_{21}}{a_{11} a_{22} - a_{12} a_{21}} = \frac{D_2}{D}$$

$$x_1 = \frac{D_1}{D} = \frac{\begin{vmatrix} b_1 & a_{12} \\ b_2 & a_{22} \end{vmatrix}}{\begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix}}, \quad x_2 = \frac{D_2}{D} = \frac{\begin{vmatrix} a_{11} & b_1 \\ a_{21} & b_2 \end{vmatrix}}{\begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix}}.$$

注意 分母都为原方程组的系数行列式.

例1 求解二元线性方程组

$$\begin{cases} 3x_1 - 2x_2 = 12, \\ 2x_1 + x_2 = 1. \end{cases}$$

解 $D = \begin{vmatrix} 3 & -2 \\ 2 & 1 \end{vmatrix} = 3 - (-4) = 7 \neq 0,$

$$D_1 = \begin{vmatrix} 12 & -2 \\ 1 & 1 \end{vmatrix} = 14, \quad D_2 = \begin{vmatrix} 3 & 12 \\ 2 & 1 \end{vmatrix} = -21,$$

$$\therefore x_1 = \frac{D_1}{D} = \frac{14}{7} = 2, \quad x_2 = \frac{D_2}{D} = \frac{-21}{7} = -3.$$

用消元法解三元线性方程组

$$\begin{cases} a_{11}x_1 + a_{12}x_2 + a_{13}x_3 = b_1, & (1) \\ a_{21}x_1 + a_{22}x_2 + a_{23}x_3 = b_2. & (2) \\ a_{31}x_1 + a_{32}x_2 + a_{33}x_3 = b_3. & (3) \end{cases}$$

方程组解的特点：

- 分母相同，由方程组的9个系数确定
- 分子、分母都是9个数中的3个相乘后再相加减

二、三阶行列式

规律：不同行不同列的
3个元素的乘积，再冠
以正号或负号！

定义 设有9个数排成3行3列的数表

$$\begin{array}{ccc} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{array} \quad (5)$$

记

$$\begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} = a_{11}a_{22}a_{33} + a_{12}a_{23}a_{31} + a_{13}a_{21}a_{32} \quad (6) \\ - a_{11}a_{23}a_{32} - a_{12}a_{21}a_{33} - a_{13}a_{22}a_{31},$$

(6) 式称为数表 (5) 所确定的**三阶行列式**.

$$D = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}$$

· 列标
行标

不能直接套用二阶行列式的对角线法则！

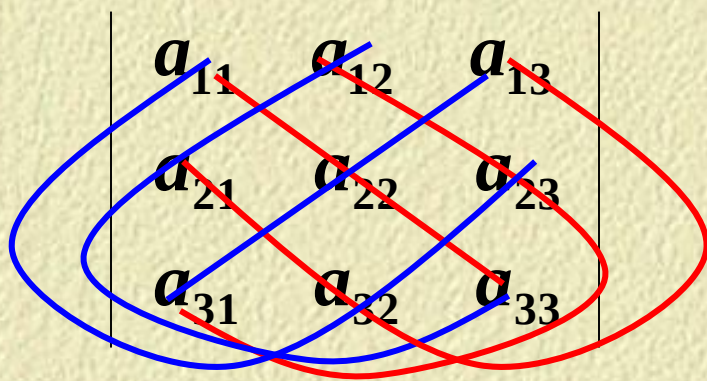
三阶行列式的计算

(1) 沙路法

Diagram illustrating the Sarrus rule for a 3x3 determinant. The first three columns are repeated to the right. Blue arrows show the positive terms (downward diagonals), and red arrows show the negative terms (upward diagonals).

$$D = a_{11}a_{22}a_{33} + a_{12}a_{23}a_{31} + a_{13}a_{21}a_{32} - a_{11}a_{23}a_{32} - a_{12}a_{21}a_{33} - a_{13}a_{22}a_{31}.$$

(2) 对角线法则



规律：不同行不同列的3个元素的乘积，再冠以正号或负号！

$$= a_{11}a_{22}a_{33} + a_{12}a_{23}a_{31} + a_{13}a_{21}a_{32} \\ - a_{13}a_{22}a_{31} - a_{12}a_{21}a_{33} - a_{11}a_{23}a_{32}.$$

注意 红线上三元素的乘积冠以正号，蓝线上三元素的乘积冠以负号。

【说明】

1. 三阶行列式包括3!项，每一项都是位于不同行，不同列的三个元素的乘积，其中三项为正，三项为负。
2. 对角线法则只适用于二阶与三阶行列式。

例2 计算三阶行列式 $D = \begin{vmatrix} 1 & 2 & -4 \\ -2 & 2 & 1 \\ -3 & 4 & -2 \end{vmatrix}$

解 按对角线法则，有

$$\begin{aligned} D &= 1 \times 2 \times (-2) + 2 \times 1 \times (-3) + (-4) \times (-2) \times 4 \\ &\quad - 1 \times 1 \times 4 - 2 \times (-2) \times (-2) - (-4) \times 2 \times (-3) \\ &= -4 - 6 + 32 - 4 - 8 - 24 \\ &= -14. \end{aligned}$$

例3 求解方程 $\begin{vmatrix} 1 & 1 & 1 \\ 2 & 3 & x \\ 4 & 9 & x^2 \end{vmatrix} = 0.$

解 方程左端

$$\begin{aligned} D &= 3x^2 + 4x + 18 - 9x - 2x^2 - 12 \\ &= x^2 - 5x + 6, \end{aligned}$$

由 $x^2 - 5x + 6 = 0$ 解得

$$x = 2 \text{ 或 } x = 3.$$

利用三阶行列式求解三元线性方程组

如果三元线性方程组
$$\begin{cases} a_{11}x_1 + a_{12}x_2 + a_{13}x_3 = b_1, \\ a_{21}x_1 + a_{22}x_2 + a_{23}x_3 = b_2, \\ a_{31}x_1 + a_{32}x_2 + a_{33}x_3 = b_3; \end{cases}$$

的系数行列式 $D = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} \neq 0,$

$x_i = \frac{D_i}{D}$ 可推广到三元线性方程组的求解上!

$$\begin{cases} a_{11}x_1 + a_{12}x_2 + a_{13}x_3 = b_1, \\ a_{21}x_1 + a_{22}x_2 + a_{23}x_3 = b_2, \\ a_{31}x_1 + a_{32}x_2 + a_{33}x_3 = b_3; \end{cases}$$

若记

$$D_1 = \begin{vmatrix} b_1 & a_{12} & a_{13} \\ b_2 & a_{22} & a_{23} \\ b_3 & a_{32} & a_{33} \end{vmatrix},$$

或

$$\begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix} D = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}$$

$$\begin{cases} a_{11}x_1 + a_{12}x_2 + a_{13}x_3 = b_1, \\ a_{21}x_1 + a_{22}x_2 + a_{23}x_3 = b_2, \\ a_{31}x_1 + a_{32}x_2 + a_{33}x_3 = b_3; \end{cases}$$

若记

$$D_1 = \begin{vmatrix} b_1 & a_{12} & a_{13} \\ b_2 & a_{22} & a_{23} \\ b_3 & a_{32} & a_{33} \end{vmatrix},$$

或

$$D_1 = \begin{vmatrix} b_1 & a_{12} & a_{13} \\ b_2 & a_{22} & a_{23} \\ b_3 & a_{32} & a_{33} \end{vmatrix},$$

$$\begin{cases} a_{11}x_1 + a_{12}x_2 + a_{13}x_3 = b_1, \\ a_{21}x_1 + a_{22}x_2 + a_{23}x_3 = b_2, \\ a_{31}x_1 + a_{32}x_2 + a_{33}x_3 = b_3; \end{cases}$$


$$D = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}$$

$$\begin{cases} a_{11}x_1 + a_{12}x_2 + a_{13}x_3 = b_1, \\ a_{21}x_1 + a_{22}x_2 + a_{23}x_3 = b_2, \\ a_{31}x_1 + a_{32}x_2 + a_{33}x_3 = b_3; \end{cases}$$

得 $D_2 = \begin{vmatrix} a_{11} & b_1 & a_{13} \\ a_{21} & b_2 & a_{23} \\ a_{31} & b_3 & a_{33} \end{vmatrix},$

$$\begin{cases} a_{11}x_1 + a_{12}x_2 + a_{13}x_3 = b_1, \\ a_{21}x_1 + a_{22}x_2 + a_{23}x_3 = b_2, \\ a_{31}x_1 + a_{32}x_2 + a_{33}x_3 = b_3; \end{cases}$$

$$D = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}$$

$$\begin{cases} a_{11}x_1 + a_{12}x_2 + a_{13}x_3 = b_1, \\ a_{21}x_1 + a_{22}x_2 + a_{23}x_3 = b_2, \\ a_{31}x_1 + a_{32}x_2 + a_{33}x_3 = b_3; \end{cases}$$

得 $D_2 = \begin{vmatrix} a_{11} & b_1 & a_{13} \\ a_{21} & b_2 & a_{23} \\ a_{31} & b_3 & a_{33} \end{vmatrix},$

$$\begin{cases} a_{11}x_1 + a_{12}x_2 + a_{13}x_3 = b_1, \\ a_{21}x_1 + a_{22}x_2 + a_{23}x_3 = b_2, \\ a_{31}x_1 + a_{32}x_2 + a_{33}x_3 = b_3; \end{cases} \Rightarrow D_3 = \begin{vmatrix} a_{11} & a_{12} & b_1 \\ a_{21} & a_{22} & b_2 \\ a_{31} & a_{32} & b_3 \end{vmatrix}.$$

$$D = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} \quad D_1 = \begin{vmatrix} b_1 & a_{12} & a_{13} \\ b_2 & a_{22} & a_{23} \\ b_3 & a_{32} & a_{33} \end{vmatrix},$$

$$D_2 = \begin{vmatrix} a_{11} & b_1 & a_{13} \\ a_{21} & b_2 & a_{23} \\ a_{31} & b_3 & a_{33} \end{vmatrix}, \quad D_3 = \begin{vmatrix} a_{11} & a_{12} & b_1 \\ a_{21} & a_{22} & b_2 \\ a_{31} & a_{32} & b_3 \end{vmatrix}.$$

则三元线性方程组的解为:

$$x_1 = \frac{D_1}{D}, \quad x_2 = \frac{D_2}{D}, \quad x_3 = \frac{D_3}{D}.$$

例4 解线性方程组

$$\begin{cases} x_1 - 2x_2 + x_3 = -2, \\ 2x_1 + x_2 + -3x_3 = 1, \\ -x_1 + x_2 - x_3 = 0. \end{cases}$$

解 由于方程组的系数行列式

$$D = \begin{vmatrix} 1 & -2 & 1 \\ 2 & 1 & -3 \\ -1 & 1 & -1 \end{vmatrix} = 1 \times 1 \times (-1) + (-2) \times (-3) \times (-1) \\ + 1 \times 2 \times 1 - 1 \times 1 \times (-1) - (-2) \times 2 \times (-1) - 1 \times (-3) \times 1 \\ = -5 \neq 0,$$

$$\begin{cases} x_1 - 2x_2 + x_3 = -2, \\ 2x_1 + x_2 + -3x_3 = 1, \\ -x_1 + x_2 - x_3 = 0. \end{cases} \quad D = \begin{vmatrix} 1 & -2 & 1 \\ 2 & 1 & -3 \\ -1 & 1 & -1 \end{vmatrix} = -5$$

同理可得 $D_1 = \begin{vmatrix} -2 & -2 & 1 \\ 1 & 1 & -3 \\ 0 & 1 & -1 \end{vmatrix} = -5,$

$$D_2 = \begin{vmatrix} 1 & -2 & 1 \\ 2 & 1 & -3 \\ -1 & 0 & -1 \end{vmatrix} = -10, \quad D_3 = \begin{vmatrix} 1 & -2 & -2 \\ 2 & 1 & 1 \\ -1 & 1 & 0 \end{vmatrix} = -5,$$

故方程组的解为:

$$x_1 = \frac{D_1}{D} = 1, \quad x_2 = \frac{D_2}{D} = 2, \quad x_3 = \frac{D_3}{D} = 1.$$

四阶行列式与n阶行列式的定义

$$\begin{vmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \\ a_{41} & a_{42} & a_{43} & a_{44} \end{vmatrix}$$

规律：不同行不同列的4个元素的乘积，再冠以正号或负号！

对角线法不适用！

四阶行列式举例

三阶行列式的沙路法和对角线法不适用四阶行列式

The diagram shows a 4x4 matrix with elements:

$$\begin{vmatrix} 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \\ 1 & 0 & 0 & 1 \end{vmatrix}$$

A red dotted line represents the main diagonal from top-left to bottom-right. Green loops connect elements in a path that would be used for a 3x3 determinant calculation (e.g., 1 to 1 to 1 to 1). To the right of the matrix, the text "=1" is followed by a blue "+1" inside a red circle, and a large red "X" is placed below it, indicating that the result is incorrect.

【例】

$$\underline{5x_1 + 4x_3 + 2x_4 = 3} \quad (1)$$

$$x_1 - x_2 + 2x_3 + x_4 = 1 \quad (2)$$

$$4x_1 + x_2 + 2x_3 = 1 \quad (3)$$

求 $x_4 = ?$

$x_i = \frac{D_i}{D}$ 可推广到n元线性方程组的求解上!

——克拉默法则 (2.4节)

由(2)+(3)

得: $x_4 = 1$

$$D = \begin{vmatrix} 5 & 0 & 4 & 2 \\ 1 & -1 & 2 & 1 \\ 4 & 1 & 2 & 0 \\ 1 & 1 & 1 & 1 \end{vmatrix}$$

$$= -7$$

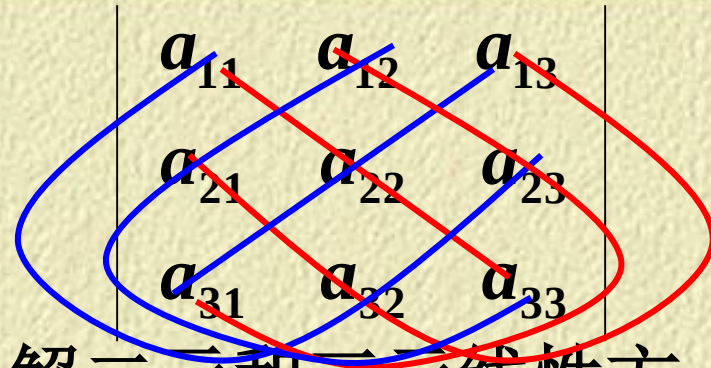
$$D_4 =$$

$$\begin{vmatrix} 5 & 0 & 4 & 3 \\ 1 & -1 & 2 & 1 \\ 4 & 1 & 2 & 1 \\ 1 & 1 & 1 & 0 \end{vmatrix}$$

$$= -7$$

$$\frac{D_4}{D} = 1$$

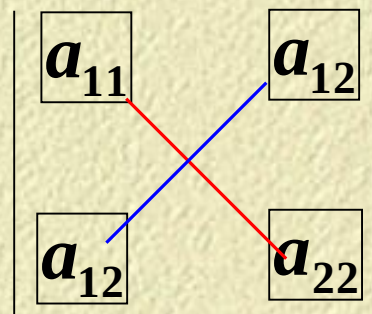
三、小结



二阶和三阶行列式是由解二元和三元线性方程组引入的.

二阶与三阶行列式的计算 —— 对角线法则

$$\begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} = a_{11}a_{22} - a_{12}a_{21}.$$



$$\begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} = a_{11}a_{22}a_{33} + a_{12}a_{23}a_{31} + a_{13}a_{21}a_{32} \\ - a_{11}a_{23}a_{32} - a_{12}a_{21}a_{33} - a_{13}a_{22}a_{31},$$

利用二阶行列式求解二元线性方程组

如果二元线性方程组
$$\begin{cases} a_{11}x_1 + a_{12}x_2 = b_1, \\ a_{21}x_1 + a_{22}x_2 = b_2. \end{cases}$$

的系数行列式 $D \neq 0$ ，则方程组的解为

$$x_1 = \frac{D_1}{D} = \frac{\begin{vmatrix} b_1 & a_{12} \\ b_2 & a_{22} \end{vmatrix}}{\begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix}}, \quad x_2 = \frac{D_2}{D} = \frac{\begin{vmatrix} a_{11} & b_1 \\ a_{21} & b_2 \end{vmatrix}}{\begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix}}.$$

利用三阶行列式求解三元线性方程组

如果三元线性方程组
$$\begin{cases} a_{11}x_1 + a_{12}x_2 + a_{13}x_3 = b_1, \\ a_{21}x_1 + a_{22}x_2 + a_{23}x_3 = b_2, \\ a_{31}x_1 + a_{32}x_2 + a_{33}x_3 = b_3; \end{cases}$$

的系数行列式
$$D = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} \neq 0,$$

则三元线性方程组的解为:

$$x_1 = \frac{D_1}{D}, \quad x_2 = \frac{D_2}{D}, \quad x_3 = \frac{D_3}{D}.$$

思考题

求一个二次多项式 $f(x)$,使

$$f(1)=0, \quad f(2)=3, \quad f(-3)=28.$$

【提示】

设所求的二次多项式为

$$f(x)=ax^2+bx+c,$$

思考题解答

解 设所求的二次多项式为

$$f(x) = ax^2 + bx + c,$$

由题意得 $f(1) = a + b + c = 0,$

$$f(2) = 4a + 2b + c = 3, \quad f(-3) = 9a - 3b + c = 28,$$

得一个关于未知数 a, b, c 的线性方程组,

$$\begin{cases} a + b + c = 0 \\ 4a + 2b + c = 3 \\ 9a - 3b + c = 28 \end{cases}$$

关于未知数a, b, c的线性方程组

$$\begin{cases} a + b + c = 0 \\ 4a + 2b + c = 3 \\ 9a - 3b + c = 28 \end{cases}$$

又 $D = -20 \neq 0$, $D_1 = -40$, $D_2 = 60$, $D_3 = -20$.

得 $a = D_1/D = 2$, $b = D_2/D = -3$, $c = D_3/D = 1$

故所求多项式为

$$f(x) = 2x^2 + -3x + 1.$$

作业

• P21 1 (3、4)

习 题 一

1. 利用对角线法则计算下列三阶行列式：

$$(1) \begin{vmatrix} 2 & 0 & 1 \\ 1 & -4 & -1 \\ -1 & 8 & 3 \end{vmatrix};$$

$$(3) \checkmark \begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ a^2 & b^2 & c^2 \end{vmatrix};$$

$$(2) \begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix};$$

$$(4) \checkmark \begin{vmatrix} x & y & x+y \\ y & x+y & x \\ x+y & x & y \end{vmatrix}.$$